Figure 8.—Model of (*a*) the helium atom; (*b*) the lithium atom.

The third period, which begins with sodium and ends with argon, again consists of eight elements. Thereafter, the periodic system becomes a little more complicated. Both the fourth period (from potassium to krypton) and the fifth one (from rubidium to xenon) contain eighteen elements each, and the sixth period (from caesium to the inert gas radon) consists of thirty-two elements. The final period is evidently an incomplete one, which for the time being ends with curium. The numbers of elements contained in the various periods—2, 8, 18, 32, i.e. 2×1^2, 2×2^2, 2×3^2, 2×4^2—obviously follow a simple mathematical sequence. The latter can be explained by the quantum numbers of the individual states of vibration, already mentioned. But we cannot discuss the details of this explanation here.

We have now obtained a general, although merely superficial, survey of our current knowledge of the extranuclear atomic structure. This knowledge enables the physicist to understand the chemical properties of the individual elements, in their general outlines at least. Theoretically, with the aid of quantum mechanics, it would be possible to calculate, quantitatively, every chemical magnitude, such as emission of heat, affinities, etc. But the mathematical difficulties are, as a rule, so great that such calculations have only been actually carried out in some of the simplest of cases.

This brings us to the conclusion of our discussions of the extranuclear structure of the atom. We shall now turn our attention to our principal topic, the nucleus of the atom.